

Designing For Sustainable and Equitable Agricultural Labor and Technologies

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ABSTRACT

This dissertation project begins from the literature-informed premise that dominant and historical conditions which have shaped the agricultural industry, namely scientific agriculture and racialized systems of migrant labor, may be perpetuated by today's commercial agtech. Implications for commercial agtech carrying these dominant values are that emerging agtech may continue to exploit racialized labor and exclude sustainable farming priorities. My research investigates how these labor production systems, which have historically been used for commercial farming practices, may be redirected to support more sustainable and socially equitable farming. My project is situated between Ontario, Canada, where the largest concentration of migrant farmers work and Oaxaca, Mexico, the home state of many migrant farmers employed in Ontario. I rely upon qualitative and design research methods to explore how migrant farmers are affected by agtech, how migrant farmers' agricultural knowledge and expertise are interpreted and used transnationally, and how Canadian growers and agtech designers envision how migrant labor fits into alternative food system futures.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI; • **Applied computing** → Agriculture.

KEYWORDS

digital agriculture, agriculture, farming, food systems, sustainability, environmental justice, climate change, migration, migrant labor

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1 INTRODUCTION

My research investigates the social, political, and ecological circumstances of how migrant farm labor and agricultural technologies (agtech) shape large-scale commercial farming practices and interrogates how these labor production systems can be redirected

to support more sustainable and socially equitable agricultural practices. I rely upon qualitative and design research methods to explore how migrant farm workers are affected by agtech, how migrant farmers' agricultural knowledge and expertise are interpreted and used transnationally, and how Canadian growers and agtech designers envision how migrant labor fits into alternative food system futures. My project is situated between Ontario, Canada and Oaxaca, Mexico as Ontario holds the largest concentration of migrant workers in Canada [26] and Oaxaca is the home state of many Mexican migrant farmers in Canada and its primary industry is agriculture [7, 13]. I link the farm sites in Ontario and Oaxaca through visiting farms that migrant interviewees worked or grew up on. I examine these contexts with attention to local knowledge practices, tools, and ecological relations with the aim of repositioning migrant farmer skillsets toward more sustainable alternative farming practices in Canada. Through my research, I also hope to redirect the design and implementation of agtech in ways that are more inclusive of migrant farmer needs, values and skillsets and that also support more sustainable, localized agricultural practices.

1.1 Context and Motivation

Since the 19th century, Canadian and American agricultural industries have dealt with domestic labor shortages [4]. Both countries have supplemented this deficit through racialized temporary workforces and the development of cost and labor-saving agricultural technologies [23, 30]. In Canada, temporary workforces have been institutionalized through the SAWP (Seasonal Agricultural Worker Program) and the TFWP (Temporary Foreign Worker Program), which developed in the 1960s and 70s [3]. The Canadian government designed these programs so that people from the Caribbean and Mexico could work in Canada temporarily under labor conditions below national wage and equity standards [31]. American agricultural science institutes, developed post-WWI, greatly influenced (and continue to influence) the design direction of agtech in Canada [30]. Common agtech include RFID tracking systems, robotic harvesters, IoT sensors for weather and irrigation, computer vision systems, unmanned aerial vehicles and human-assistive technologies such as robotic carts [28]. Temporary labor programs and most agtech serve the demands of large-scale commercial agricultural initiatives, which are primarily driven by profitability and efficiency [30].

In contrast to agriculture industrialization, alternative agricultural movements such as Fair Trade, food sovereignty, and organic farming initiatives work together to restructure local and global food systems to become more sustainable and socially equitable [5]. These movements support small-scale and subsistence farming in the Global North and South, leaving open the question of how these

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practices can scale to serve urban populations [5]. Furthermore, alternative food movements rarely consider temporary workforces nor technology's role in supporting sustainable farming [6]. My research begins from the premise that dominant and historical conditions which have shaped the agricultural industry, namely scientific agriculture, and racialized systems of migrant labor, may be perpetuated in today's commercial agtech design decisions [30]. Implications for commercial agtech carrying these dominant values are that emerging agtech may continue to exploit racialized labor and exclude alternative agricultural priorities.

1.2 Key Related Work

I am situating my dissertation in both HCI and critical social science scholarship because I see the need to contribute to socially responsible and sustainable material design of agtech while simultaneously supporting accompanying labor policies regarding technologies. HCI literature on agriculture is dispersed across the subdomains of Rural Computing [16], Information and Community Technology for Development (ICTD) [27], Community Informatics [17], and Sustainable HCI (SHCI) [33]. Each subdomain holds specific epistemological commitments that influence how researchers engage with participants and create sociotechnical outcomes. For example, ICTD projects mainly occur in the Global South and focus on the needs of subsistence farmers. Many ICTD projects support subsistence farmers to improve their livelihoods by designing information communication technologies that bridge knowledge gaps and enable communication with extension officers [27]. SHCI scholarship, meanwhile, calls for researchers to reduce material effects of technology, improve ways for maintenance, and encourage people to behave and manage environments more sustainably [19]. SHCI projects are typically in the Global North and focus on small-scale organic farmers [25]. Common sociotechnical outcomes for SHCI projects are design recommendations [24], information systems [25] and design prototypes that aim to strengthen human-nature relations [22]. Overall, HCI projects that address agriculture are geographically dispersed, varied in farm type, and disparate in values and missions regarding how to support agricultural actors and practices.

Critical social science digital agriculture projects are generally concentrated on farm sites in the Global North and focus on small-scale farmers as primary social actors. Critical social scientists often adopt the STS epistemological perspective that technologies are not neutral artefacts but value-laden with the qualities and objectives that engineers, designers and stakeholders ascribe to them [37]. These researchers thus begin with an underlying criticality toward technology and its current role in agricultural environments. Scholars focus on such themes as technological adoption by farmers, farmer resistance and agency and how farmer knowledge can be better incorporated into agtech [30]. Most critical social science scholarship share the viewpoint that the trajectory for agtech currently aligns with the dominant vision for our food systems, which is rooted in agricultural industrialization. Scholars work to counteract this dominant trajectory by supporting policy development and frameworks that promote inclusive and sustainable governance of agtech [29].

I have identified five ways through which I hope to bridge HCI and critical social science agriculture-related scholarship and critical knowledge gaps through my dissertation. First, my project addresses the impacts of agtech on migrant farmers in Canada. Farm workers are rarely the focal point in existing scholarship. Second, my investigation of migrant farmers' practices in Ontario and Mexico enables me to explore transnational entanglements between the Global North and South. This inquiry crosses national and developmental boundaries, blurring definitions of what constitutes equitable labor. Third, I hope to contribute to SHCI and sustainability-related social science literature by considering sustainability alongside environmental justice concerns. I see my work as a way to expand current SHCI priorities, which offer very little attention to issues of social equity. Fourth, building on the work of Liu and Sengers [23], I plan to historically situate my work in how dominant agricultural production and labor systems have been established. These histories are critical to understanding what values, assumptions and social actors have shaped current agricultural processes and tools. Finally, I intend to bridge HCI and critical social science by creating material design interventions that support the development of sustainable and equitable tech.

2 THEORETICAL APPROACHES

There are three theoretical areas that I plan to draw from to frame my research: (1) Science and Technology Studies (STS) and feminist technoscience, (2) postcolonial and decolonial computing and (3) environmental justice. These approaches are well-suited to investigate the politics between agricultural innovation, migrant farmers, and alternative sustainable food movements. I describe how I plan to employ these approaches below.

2.1 STS and Feminist Technoscience

I will adopt an STS and feminist technoscience approach in my research design by maintaining the position that technologies are value-laden and shaped by dominant political, social, and economic systems [36]. Embedded in this viewpoint is the understanding that technologies, individuals, and organisations are shaped by societal structures and power relations. Following feminist technoscience scholars, Harding [15] and Haraway [14], I will also maintain that knowledges must be situated in local contexts, and these knowledges are partial, multiple, and often conflicting. My research objective of studying migrant farmer experiences and knowledges both in Ontario and Oaxaca is an example of how I am committed to exploring how knowledges are interpreted, used, and reproduced across local environments.

2.2 Postcolonial and Decolonial Computing

In HCI, postcolonial and decolonial standpoints emphasize how all design research and technological artifacts are culturally situated and power laden [18]. Dourish and Mainwaring [10] illustrate how ubiquitous computing carries colonial values such as universality and how 'progress' is defined through innovations and policies established in the Global North. Decolonial scholars such as Ali [1] and Lazem et al. [20] argue that postcolonial computing does not go far enough in its mandate: the movement's prioritization of cultural issues over political and economic concerns cause a lack

of focus on material issues. As an alternative, Ali [1] proposes a decolonial stance to computing, recommending scholars to study who is speaking and from where. My research questions will benefit from these approaches as I investigate how migrant worker knowledges are interpreted and exploited between the Global North and South. By raising the historical origins of temporary programs and following the body and geo-politics of knowledge, I will investigate how colonial relations have shaped temporary agricultural labor and related technologies in Canada.

2.3 Environmental Justice

Environmental justice seeks to equalise distribution of toxicities and waste and increase recognition and representation for communities particularly susceptible to environmental harms [12]. Environmental justice scholars such as Shadaan and Murphy [32], Liboiron [21], and Vasudevan and Smith [35] aim to increase the legal recognition and political representation of such communities to help them gain environmental reparations. I rely upon an environmental justice lens to frame my design work, which jointly aims to increase legal and political representation of migrant workers' hopes, concerns and needs regarding their livelihoods as well as create material design interventions that shift agtech design and the marketing of food commodities toward more environmentally justice agricultural labor practices.

3 RESEARCH QUESTIONS

My research project addresses the following interrelated questions, drawing on fieldwork across Canada and Mexico:

- (1) How are migrant farmers impacted by commercial agtech in Canada?
- (2) How may migrant farmers operate within emerging visions and technologies that support socially equitable and sustainable food systems?
- (3) What design interventions might we perform to connect sustainability and equitable forms of migrant labor?

4 METHODS, RATIONALE, AND RESEARCH PROGRESS TO DATE

To answer these questions, I have four planned research phases: three phases of qualitative fieldwork with migrant farmers and other agriculture actors in Canada and Mexico followed by a fourth phase of participatory design workshops. All research with migrant farmers is expected to be conducted in Spanish by either myself or a Spanish-speaking member of community organization that support migrant farmers. The first phase begins with an exploratory survey and semi-structured interviews with migrant farmers in Ontario. With little existing research on the topic, the survey is a critical step to determine which agricultural sectors and tasks workers perceive as most affected by technologies. So far, 67 respondents out of an intended 100 have completed the survey. I am also conducting 15-20 semi-structured interviews with selected survey participants to gain a richer perspective of their work experiences and perceptions of equitable labor and technology. I have conducted 10 of these interviews so far.

The second phase will take place in Oaxaca in February 2023, where I plan to visit 6-8 subsistence farmers at their farm sites and

conduct semi-structured interviews. I will focus on labor processes, tools as well as farmer perspectives on required agricultural expertise and knowledge. Conversations with migrant workers will direct my selection of farm sites in Oaxaca. When possible, I will visit the family farms of the migrant workers interviewed in Ontario. This research is not an attempt at forming a one-to-one comparison of farm sites as the farms in Mexico will be subsistence rather than commercial farms. Furthermore, farm workers in Canada may not be representative of the farmers I encounter in Mexico. Instead, the aim of this investigation is to better understand how migrant farmers' professional skill sets and experiences are interpreted and utilized in both Canada and in Mexico.

The third phase will take place in Summer 2023. I will conduct 5-10 semi-structured interviews and field site visits with small-scale farmers in Ontario. I will also interview 5-10 technologists and researchers in agtech. The intention of this phase is to understand agriculture stakeholder perspectives on how farm labor is managed, how they imagine how migrant labor will fit into alternative food system futures and how they consider sustainability and social equity in agtech design decisions.

The data collected through the survey, interviews, field site visits and participant observations will inform the fourth phase of my project, which will consist of three participatory design workshops. As these workshops will be informed through empirical data, I do not know yet the precise aims of the sessions, but I anticipate that they will include migrant farmers, community representatives, small-scale growers and agtech designers. As I expect to leverage speculative design and critical making practices, some potential outcomes may be how migrant farmers envision their future employment and relationship to Canada, value tensions, needs and assumptions that exist between actors, and how technologies may be designed, repurposed or replaced to support these desired values and pathways. I particularly want to understand how stakeholders define and envision equitable labor and sustainability with the hopes of exposing differences but also establishing common understandings of these concepts. Using the empirical evidence described above, I seek to improve current agtech development through material design interventions.

5 RESULTS AND CONTRIBUTIONS TO DATE

My academic contributions so far have consisted of literature reviews and one design project. I have presented a preliminary literature review of HCI agriculture scholarship in a *CHI 2021* workshop focused on migration [9], and organized a workshop on essential labor spheres and the impacts of automation for *CHI 2022* [11]. I also published a speculative design project where I used mushroom package labels as ways of showcasing different 'realities' of labor production [8]. By considering what mushroom labels would look like if they emphasized migrant labor conditions or automation over other information such as organic production standards, I intended to bring hidden labor inequities closer to end-consumers. I also presented at the 2021 *Society for Social Studies of Science (4S)* conference on preliminary findings regarding migrant labor and agtech. I am currently revising a conference paper for *CHI 2023* as the first author, which is a literature review of agriculture-related ACM scholarship through the lens of competing future visions of

food systems. I am expected to present early findings from my first research phase at the 4S conference in December 2022.

Based on preliminary findings from surveys and interviews completed, I have identified that temporary greenhouse workers are the most affected by agtech. The most common agtech mentioned are productivity tracking technologies, or as workers call them, *las ponchadoras*, which monitor workers' performance through RFID or biometric sensors. Workers also report interactions with robots, irrigation sensors and technologies that dispense pesticides. Other early results include: there are unexpectedly high numbers of laborers working illegally and this illegal status drives the kind of work that people do and the technologies that they interact with; workers' perceptions of technologies appear to be both skeptical and hopeful – on one end, workers believe technologies will only support employer objectives and on the other, workers are hopeful technologies will reduce the physical strain of their work; and finally, looking ahead 10 years, most workers do not believe their jobs will be replaced.

6 OPEN QUESTIONS AND CHALLENGES

I see the *CHI 2023* doctoral consortium as a way to think through the first two phases of my field work and plan for the following research and design stages. Below I list my most pressing questions and challenges that I face in my dissertation, which I would like to discuss with consortium participants.

- (1) I see my research as interdisciplinary between HCI, STS and environmental studies. I am still figuring out how to situate my research in HCI. Do other students have similar disciplinary concerns? How do they address them?
- (2) How do others conduct, connect, and compare field work across international field sites?
- (3) From preliminary findings I have identified that migrant workers generally do not want to interact with surveillance technologies and rather they not exist. What if one of my outcomes is not to create more technologies but to recommend the deconstruction of certain technologies that are harmful for workers? Where do participants think this type of deconstructive work fits into HCI scholarship, which often focuses on more generative acts of design?
- (4) Bardzell et al. recently argued that “agriculture requires intervention” [2], particularly interventions that support interspecies relations and that build shareable, place-based design practices. How can I extend these SHCI provocations for intervention to include issues of environmental justice such as those related to equitable labor?
- (5) Material design interventions seek to shift or change the status quo rather than simply supporting pre-existing processes or infrastructures. Are other students intending to use their research to develop interventions and if so, what do their interventions look like and how are they positioning their work within the field of HCI?
- (6) I want to avoid damage-centred research approaches, which document the pain and loss of communities and individuals “to obtain particular political or material gains” [34]. How can I avoid damage-centered approaches while still

supporting community organizations that often rely upon these framings to drive policy development?

- (7) The outcomes of my research will emerge based on what I learn from empirical data collected. How do other students manage when their research directions and outcomes change based on their empirical findings?

7 DISSERTATION STATUS AND LONG-TERM GOALS

I am a third year PhD candidate at the Faculty of Information and School of Environment at the University of Toronto. My dissertation research is in progress: I have received ethics approval for the first two research phases, and am collecting data for the exploratory survey and field work with migrant farmers in Ontario. By the time of the 2023 *CHI* conference, I expect to have completed data collection and translation for the first two phases and be organizing the final research phase and participatory design workshops. I anticipate finishing my field work by Spring 2024 and defending my dissertation by Spring 2025. Around the time of my defence, I will be on the job market hoping to obtain a research and teaching position at an academic institution. Ideally, I would like to obtain a position that is cross-disciplinary between environmental studies and information science.

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